

where $\alpha > 0$ and $\beta \geq 1$. The experiments use fixed α and sweep β to vary the convexity of congestion costs.

An origin-destination pair $r = (o, d)$ has demand D_r . For each OD pair, a route-generation procedure returns a finite candidate set $\mathcal{P}_r$. A recommendation policy maps $(G, r, \mathcal{P}_r, D_r)$ and a random seed, if any, to route flows $f_{rP} \geq 0$ satisfying $\sum_{P \in \mathcal{P}_r} f_{rP} = D_r$. Edge flows are induced by route flows:

$$x_e = \sum_r \sum_{P \in \mathcal{P}_r: e \in P} f_{rP}.$$

The social cost of an assignment is

$$C(f) = \sum_{e \in E} x_e \ell_e(x_e).$$

Average realized latency is $C(f)/\sum_r D_r$.

Symbol or metric	Meaning in this paper
D_r	Demand for OD pair r .
$\mathcal{P}_r$	Finite candidate routes returned by the route-set construction procedure.
f_{rP}	Flow on route P for OD pair r .
x_e	Total edge flow induced by all OD route flows.
$\ell_e(x_e)$	Congestion-sensitive edge latency.
$C(f)$	Total system cost, equal to flow-weighted latency.
ρ	Shared-noise parameter; larger values make user recommendations more correlated.
q	Compliance rate with the shared monoculture recommendation.
HHI	Route concentration index; for one OD pair, $\sum_{P \in \mathcal{P}_r} (f_{rP}/D_r)^2$.
PoM	Empirical price of monoculture against the best implemented diversified baseline.

Recommendation policies. A deterministic monoculture policy gives all users for an OD pair the same route. The canonical monoculture baseline ranks candidate routes by free-flow travel time and assigns the full OD demand to the shortest candidate. A shared noisy monoculture samples one perturbed edge-weight model and then assigns all users according to the resulting shared ranking.

A diversified policy splits demand across candidates. The implementation includes uniform top- k splitting, softmax splitting by free-flow route cost, multiple independently perturbed recommenders, random route choice, partial-compliance variants, and a greedy marginal-social-cost baseline. The greedy policy assigns users sequentially to the route with the smallest current marginal increase in total social cost; it is a tractable oracle-style baseline, not a certified optimum.

Correlation and partial compliance. The correlated-recommender family interpolates between independent recommender noise and a shared ranking. A parameter $\rho \in [0, 1]$ controls the degree of shared perturbation. Higher ρ means users see more correlated route scores. Partial compliance is controlled by $q \in [0, 1]$: a q fraction follows the shared monoculture recommendation, while the remaining $1 - q$ fraction follows a diversified fallback.

These two parameters separate two effects that are often conflated. The parameter ρ describes correlation in the recommendation model itself: users may be exposed to similar rankings even before they choose whether to follow them. The parameter q describes behavioral uptake of a shared recommendation. A high- ρ , low- q regime can be mostly harmless if many users ignore the shared advice. A low- ρ , high- q regime can still diversify demand if users receive meaningfully different route rankings.

Empirical Price of Algorithmic Monoculture. For a tested instance I , define

$$\text{PoM}(I) = \frac{C(f_I^{\text{mono}})}{\min\{C(f_I^{\text{div}}) : f_I^{\text{div}} \in \mathcal{B}_I\}},$$

where $\mathcal{B}_I$ is the set of implemented diversified and oracle-style baselines available for that instance. This is an empirical comparison over a finite benchmark suite. It should not be read as a universal upper or lower bound, and it can be near 1 or below 1 in regimes where the free-flow-shortest route is also capacity-compatible.

Relation to price of anarchy. The notation PoM is intentionally separate from the classical price of anarchy. Price-of-anarchy analyses compare equilibrium outcomes to optima under a specified game form. Here the numerator is not necessarily an equilibrium; it is the outcome of a shared recommendation rule that correlates users onto the same route. The denominator is also not a mathematical optimum; it is the best implemented diversified baseline in the benchmark. Thus PoM is a diagnostic for recommendation-induced concentration, not a replacement for equilibrium inefficiency.

Unit of observation. Each experimental row fixes a network generator, seed, route-set construction method, OD demand, latency exponent, and recommendation policy. Summary tables distinguish raw stress-test rows from filtered rows with bounded maximum utilization. This distinction is essential because the synthetic generator can deliberately create severe overload. The paper uses those overloads to expose the mechanism, while reserving the strongest headline claim for the maximum-utilization-at-most-2 filter.

3 Two-Link Capacity-Trap Analysis

The simplest mechanism is a two-link network with one OD pair and total demand D . Link A is a shortcut: it has lower free-flow time but lower capacity than link B :

$$t_A < t_B, \quad 0 < c_A < c_B.$$

Both links use the latency function

$$\ell_i(x) = t_i \left(1 + \alpha(x/c_i)^\beta\right), \quad i \in \{A, B\}.$$

A free-flow monoculture recommender ranks A first and sends all demand to A . Its cost is

$$C_{\text{mono}}(D) = Dt_A \left(1 + \alpha(D/c_A)^\beta\right).$$

A split assignment sends pD flow to A and $(1-p)D$ flow to B , with cost

$$C_{\text{split}}(D, p) = pDt_A \left(1 + \alpha(pD/c_A)^\beta\right) + (1-p)Dt_B \left(1 + \alpha((1-p)D/c_B)^\beta\right).$$